

Grade 7
Unit 5
Lesson 6 Practice

Name Answer Key
Date _____
Period _____

Shannon's TV, Figure D, has a height of 22 in. and width of 33 in. Shannon drew a scale model of her TV screen, Figure E, with a height of 4 in. width of 6 in. Use this information to complete problems 1-5.



1. What is the scale factor from Figure D to Figure E?

$$E \text{ is } \frac{2}{11} \text{ of } D$$

2. What is the area of Shannon's TV screen?

$$22 \cdot 33 = 726 \text{ in}^2$$

3. What is the area of Shannon's scale model?

$$4 \cdot 6 = 24 \text{ in}^2$$

4. What is the ratio of the area of Figure E to the ratio of the area of Figure D?

$$\frac{726}{24} = \frac{121}{4}$$

5. What is the relationship between the scale factor and the ratios of the areas?

The square of the scale factor is the ratio of the areas.

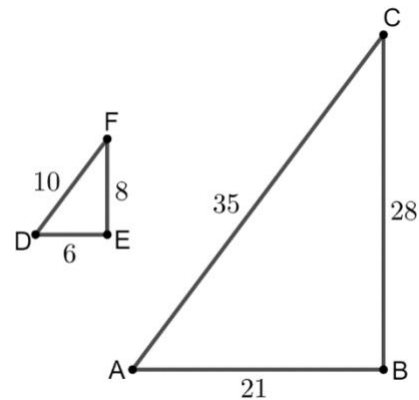
From Figure D to Figure E, the scale factor is $\frac{2}{11}$ and the ratio of the areas is $\frac{4}{121}$.

From Figure E to Figure D, the scale factor is $\frac{11}{2}$ and the ratio of the areas is $\frac{121}{4}$.

Right triangle DEF has a height of 8 units and base of 6 units. Right triangle ABC is a scaled-up image of right triangle DEF . Right Triangle ABC has a height of 28 units and base of 21 units. Use this information to complete questions 6-10.

6. Complete the table below.

	Triangle DEF	Triangle ABC
Perimeter	24 units	84 units
Area	24 units ²	294 units ²



7. What is the scale factor from Triangle DEF to Triangle ABC ?

ABC is 3.5 times larger than DEF .

8. How does the perimeter of Triangle DEF compare to Triangle ABC ?

The perimeter of ABC is 3.5 times larger than the perimeter of DEF .

9. How do the perimeters of Triangle DEF and Triangle ABC relate to the scale factor?

The scale factor, 3.5, applies not just to separate dimensions but to the perimeter as well.

10. How do the ratios of the areas of Triangle DEF and Triangle ABC relate to the scale factor?

The scale factor is squared

$$\frac{294}{24} = 12.25 = 3.5^2$$